

Lecture 9: Newton method for multi-variable unconstrained optimization problem

MA3236 Non-Linear Programming

Hu Yukuan

yukuan.hu@nus.edu.sg

S17-08-21

Quick review

- Most stationarity equations cannot be solved in closed form.
- Iterative framework: initialization + update + termination.

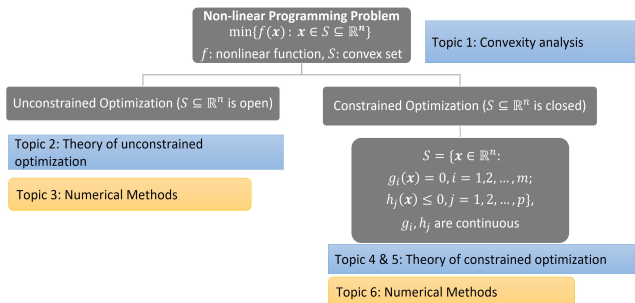
$$\mathbf{x}^{(k+1)} := \mathbf{x}^{(k)} + \alpha^{(k)} \mathbf{d}^{(k)}.$$

- Line search: one-dimensional subproblem.

$$\alpha^{(k)} \approx \arg \min_{\alpha \geq 0} f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)}).$$

- One-dimensional methods: bisection, Newton, and golden-section methods.

Topics



In Topic 3, we will study numerical methods for solving **unconstrained optimization problem**.

- Methods for solving one-variable unconstrained optimization.
- Methods for solving multi-variable unconstrained optimization: **Newton method**, steepest descent method, conjugate-gradient method

Pure Newton: backend — local quadratic model

At $\mathbf{x}^{(k)}$, let

$$\mathbf{g}^{(k)} := \nabla f(\mathbf{x}^{(k)}), \quad \mathbf{H}^{(k)} := \mathbf{H}_f(\mathbf{x}^{(k)}).$$

For a trial displacement $\mathbf{d}$, the **second-order Taylor model** is

$$m^{(k)}(\mathbf{d}) = f(\mathbf{x}^{(k)}) + (\mathbf{g}^{(k)})^T \mathbf{d} + \frac{1}{2} \mathbf{d}^T \mathbf{H}^{(k)} \mathbf{d}.$$

Its stationarity equation is

$$\nabla_{\mathbf{d}} m^{(k)}(\mathbf{d}) = \mathbf{g}^{(k)} + \mathbf{H}^{(k)} \mathbf{d} = \mathbf{0}.$$

Newton uses a solution of this linear equation as its trial step.

Pure Newton: backend — local quadratic model (cont'd)

The Newton equation is

$$\mathbf{H}^{(k)} \mathbf{d}_N^{(k)} = -\mathbf{g}^{(k)}.$$

- If $\mathbf{H}^{(k)} \succ 0$, the unique **global minimizer** of $m^{(k)}$ is

$$\mathbf{d}_N^{(k)} = -(\mathbf{H}^{(k)})^{-1} \mathbf{g}^{(k)}.$$

- If $\mathbf{H}^{(k)} \not\succ 0$, $m^{(k)}$ may not have a global minimizer / may have infinitely many global minimizers.

Pure Newton: backend — local quadratic model (cont'd)

The Newton equation is

$$\mathbf{H}^{(k)} \mathbf{d}_N^{(k)} = -\mathbf{g}^{(k)}.$$

- If $\mathbf{H}^{(k)} \succ 0$, the unique **global minimizer** of $m^{(k)}$ is

$$\mathbf{d}_N^{(k)} = -(\mathbf{H}^{(k)})^{-1} \mathbf{g}^{(k)}.$$

- If $\mathbf{H}^{(k)} \not\succ 0$, $m^{(k)}$ may not have a global minimizer / may have infinitely many global minimizers.
- If $\mathbf{H}^{(k)}$ is nonsingular, $\mathbf{d}_N^{(k)}$ is the unique **solution** to the equation.

Pure Newton: pseudocode

Choose $\mathbf{x}^{(0)} \in \mathcal{D}$, $\epsilon_g > 0$, and $K_{\max}$. For $k = 0, 1, \dots, K_{\max} - 1$:

- 1 Compute $\mathbf{g}^{(k)} = \nabla f(\mathbf{x}^{(k)})$. Stop if $\|\mathbf{g}^{(k)}\| \leq \epsilon_g$.
- 2 (Form $\mathbf{H}^{(k)} = \mathbf{H}_f(\mathbf{x}^{(k)})$ and) solve

$$\mathbf{H}^{(k)} \mathbf{d}_N^{(k)} = -\mathbf{g}^{(k)}.$$

- 3 Take the full step $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \mathbf{d}_N^{(k)}$.

Python code: `pure_newton(...)` in `109_newton_methods.py`

Pure Newton: local quadratic convergence

Let f be twice differentiable on an open neighborhood of a stationary point $\mathbf{x}^*$. Assume

- ① $\mathbf{H}_f(\mathbf{x}^*)$ is nonsingular;
- ② the Hessian is locally Lipschitz near $\mathbf{x}^*$:

$$\|\mathbf{H}_f(\mathbf{x}) - \mathbf{H}_f(\mathbf{y})\|_2 \leq L\|\mathbf{x} - \mathbf{y}\|;$$

- ③ $\mathbf{x}^{(0)}$ is sufficiently close to $\mathbf{x}^*$.

Then the iterates are **well defined** and **converge Q-quadratically**:

$$\|\mathbf{x}^{(k+1)} - \mathbf{x}^*\| \leq M\|\mathbf{x}^{(k)} - \mathbf{x}^*\|^2.$$

Pure Newton: local quadratic convergence (cont'd)

Let $\mathbf{e}^{(k)} = \mathbf{x}^{(k)} - \mathbf{x}^*$. The fundamental theorem of calculus gives

$$\nabla f(\mathbf{x}^{(k)}) = \left(\int_0^1 \mathbf{H}_f(\mathbf{x}^* + t\mathbf{e}^{(k)}) dt \right) \mathbf{e}^{(k)}.$$

Substitute this into the Newton update:

$$\mathbf{e}^{(k+1)} = (\mathbf{H}^{(k)})^{-1} \left(\int_0^1 [\mathbf{H}^{(k)} - \mathbf{H}_f(\mathbf{x}^* + t\mathbf{e}^{(k)})] dt \right) \mathbf{e}^{(k)}.$$

Bounded inverse + Lipschitz Hessian gives

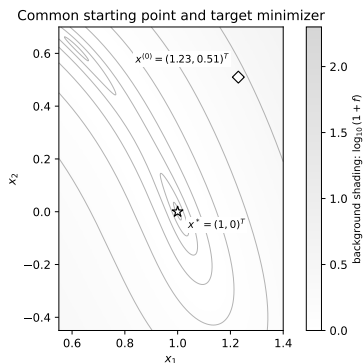
$$\|\mathbf{e}^{(k+1)}\| = O(\|\mathbf{e}^{(k)}\|^2).$$

A common test problem

A coupled-quartic test function on $\mathcal{D} = \mathbf{R}^2$:

$$f(x_1, x_2) = (x_1^2 + x_2 - 1)^2 + (x_1 + x_2^2 - 1)^2,$$

with common initial point $\mathbf{x}^{(0)} = (1.23, 0.51)^T$. Since $f \geq 0$ and $f(1, 0) = 0$, $\mathbf{x}^* = (1, 0)^T$ is a global minimizer.



Pure Newton on the common test

Selected iterates; tolerance: $\epsilon_g = 10^{-8}$.

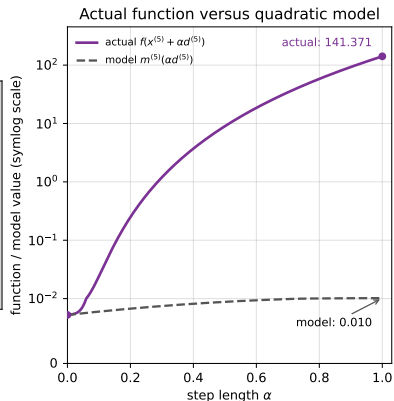
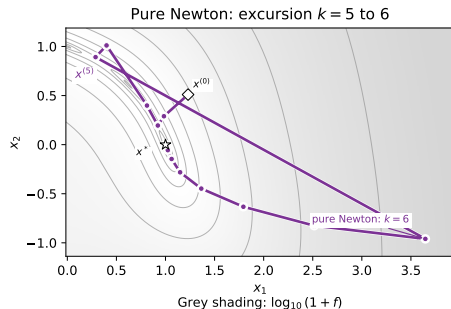
k	$x_1^{(k)}$	$x_2^{(k)}$	$f(\mathbf{x}^{(k)})$	$\ \mathbf{g}^{(k)}\ $
0	1.2300	0.5100	1.2865	6.740
2	0.9219	0.1950	0.003624	0.104
3	0.8112	0.3983	0.004089	0.139
4	0.3993	1.0107	0.2060	2.325
5	0.2877	0.8916	0.007482	0.279
6	3.6472	-0.9599	141.3709	172.831
15	1.0000	-0.0000	3.559×10^{-12}	1.998×10^{-6}
16	1.0000	-0.0000	2.201×10^{-21}	4.979×10^{-11}

Pure Newton on the common test

Selected iterates; tolerance: $\epsilon_g = 10^{-8}$.

k	$x_1^{(k)}$	$x_2^{(k)}$	$f(\mathbf{x}^{(k)})$	$\ \mathbf{g}^{(k)}\ $
0	1.2300	0.5100	1.2865	6.740
2	0.9219	0.1950	0.003624	0.104
3	0.8112	0.3983	0.004089	0.139
4	0.3993	1.0107	0.2060	2.325
5	0.2877	0.8916	0.007482	0.279
6	3.6472	-0.9599	141.3709	172.831
15	1.0000	-0.0000	3.559×10^{-12}	1.998×10^{-6}
16	1.0000	-0.0000	2.201×10^{-21}	4.979×10^{-11}

Pure Newton on the common test (cont'd)



- At $k = 5$, $\text{eig}(\mathbf{H}^{(5)}) \approx (-0.000835, 11.250)$ and $(\mathbf{g}^{(5)})^T \mathbf{d}_N^{(5)} \approx 0.005384 > 0$.
- Full step: $f(\mathbf{x}^{(5)}) \approx 0.007482$ increases to $f(\mathbf{x}^{(6)}) \approx 141.371$, while the quadratic model gives only $m^{(5)}(\mathbf{d}_N^{(5)}) \approx 0.010174$.

Pure Newton can converge to a saddle point

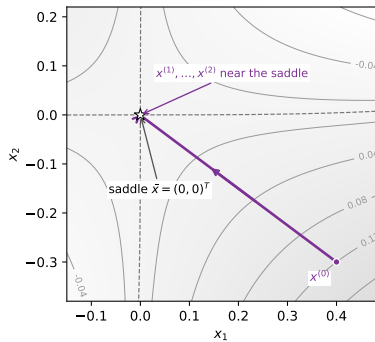
The local theorem requires nonsingularity, not positive definiteness.

$$F(x_1, x_2) = \frac{(x_1 + x_2)^4}{16} - x_1 x_2, \quad \mathbf{x}^{(0)} = (0.4, -0.3)^T.$$

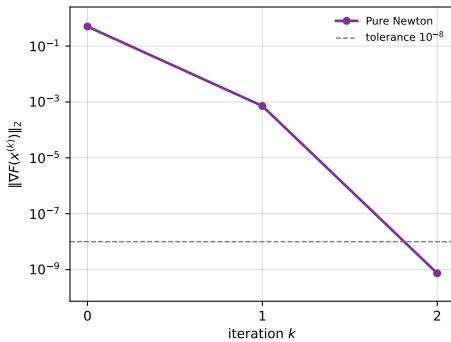
At the stationary point $\mathbf{0}$,

$$\mathbf{H}_F(\mathbf{0}) = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}, \quad \lambda = \pm 1 : \quad \mathbf{0} \text{ is a saddle point.}$$

Function contours and Newton iterates



Gradient norm



Positive definiteness implies local decrease

At a nonstationary iterate, a vector $\mathbf{d}^{(k)}$ is a **descent direction** if

$$(\mathbf{g}^{(k)})^T \mathbf{d}^{(k)} < 0.$$

Differentiability gives

$$f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)}) = f(\mathbf{x}^{(k)}) + \alpha (\mathbf{g}^{(k)})^T \mathbf{d}^{(k)} + o(\alpha),$$

so every sufficiently small $\alpha > 0$ decreases f .

Positive definiteness implies local decrease

At a nonstationary iterate, a vector $\mathbf{d}^{(k)}$ is a **descent direction** if

$$(\mathbf{g}^{(k)})^T \mathbf{d}^{(k)} < 0.$$

Differentiability gives

$$f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)}) = f(\mathbf{x}^{(k)}) + \alpha (\mathbf{g}^{(k)})^T \mathbf{d}^{(k)} + o(\alpha),$$

so every sufficiently small $\alpha > 0$ decreases f .

Newton direction

If $\mathbf{H}^{(k)} \succ 0$ and $\mathbf{H}^{(k)} \mathbf{d}_N^{(k)} = -\mathbf{g}^{(k)} \neq \mathbf{0}$, then

$$(\mathbf{g}^{(k)})^T \mathbf{d}_N^{(k)} = -(\mathbf{d}_N^{(k)})^T \mathbf{H}^{(k)} \mathbf{d}_N^{(k)} < 0.$$

What if $\mathbf{H}^{(k)} \not\succ 0$? Singular or having negative curvatures?

Shifted Newton: backend — diagonal shifting

Shifted Newton direction

If $\mathbf{B}^{(k)} := \mathbf{H}^{(k)} + \tau^{(k)}\mathbf{I} \succ 0$ for some $\tau^{(k)} \geq 0$ and $\mathbf{B}^{(k)}\mathbf{d}_M^{(k)} = -\mathbf{g}^{(k)} \neq \mathbf{0}$,

$$(\mathbf{g}^{(k)})^T \mathbf{d}_M^{(k)} = -(\mathbf{d}_M^{(k)})^T \mathbf{B}^{(k)} \mathbf{d}_M^{(k)} < 0.$$

Shifted Newton: backend — diagonal shifting

Shifted Newton direction

If $\mathbf{B}^{(k)} := \mathbf{H}^{(k)} + \tau^{(k)}\mathbf{I} \succ 0$ for some $\tau^{(k)} \geq 0$ and $\mathbf{B}^{(k)}\mathbf{d}_M^{(k)} = -\mathbf{g}^{(k)} \neq \mathbf{0}$,

$$(\mathbf{g}^{(k)})^T \mathbf{d}_M^{(k)} = -(\mathbf{d}_M^{(k)})^T \mathbf{B}^{(k)} \mathbf{d}_M^{(k)} < 0.$$

It suffices to choose $\tau^{(k)}$:

$$\tau^{(k)} + \lambda_{\min}(\mathbf{H}^{(k)}) > 0 \text{ (not necessary!)}$$

Modified quadratic model:

$$M^{(k)}(\mathbf{d}) := f(\mathbf{x}^{(k)}) + (\mathbf{g}^{(k)})^T \mathbf{d} + \frac{1}{2} \mathbf{d}^T \mathbf{B}^{(k)} \mathbf{d}.$$

Then $\mathbf{d}_M^{(k)} = -(\mathbf{B}^{(k)})^{-1} \mathbf{g}^{(k)}$ is the unique global minimizer of $M^{(k)}$.

Shifted Newton: backend — diagonal shifting (cont'd)

Shifted Newton direction

With $\mathbf{B}^{(k)} := \mathbf{H}^{(k)} + \tau^{(k)}\mathbf{I} \succ 0$,

$$\mathbf{d}_M^{(k)} = -(\mathbf{B}^{(k)})^{-1}\mathbf{g}^{(k)}.$$

For a fixed $\mathbf{H}$ and $\mathbf{g}$, let $\mathbf{d}_\tau = -(\mathbf{H} + \tau\mathbf{I})^{-1}\mathbf{g}$. Then

$$\tau\mathbf{d}_\tau = -\left(\mathbf{I} + \frac{1}{\tau}\mathbf{H}\right)^{-1}\mathbf{g} \longrightarrow -\mathbf{g} \quad \text{as } \tau \rightarrow \infty.$$

- $\tau \rightarrow 0$: Newton.
- $\tau \rightarrow \infty$: steepest descent (next lecture).

From diagonal shifting to majorization

Replace the Hessian by **any** symmetric positive-definite matrix $\mathbf{B}^{(k)}$:

$$M^{(k)}(\mathbf{d}) = f(\mathbf{x}^{(k)}) + (\mathbf{g}^{(k)})^T \mathbf{d} + \frac{1}{2} \mathbf{d}^T \mathbf{B}^{(k)} \mathbf{d}.$$

Its unique minimizer is $\mathbf{d}^{(k)} = -(\mathbf{B}^{(k)})^{-1} \mathbf{g}^{(k)}$. If a curvature bound proves

$$f(\mathbf{x}^{(k)} + \mathbf{d}) \leq M^{(k)}(\mathbf{d}) \quad \text{on a region containing } \mathbf{0} \text{ and } \mathbf{d}^{(k)},$$

then

$$f(\mathbf{x}^{(k)} + \mathbf{d}^{(k)}) \leq M^{(k)}(\mathbf{d}^{(k)}) \leq M^{(k)}(\mathbf{0}) = f(\mathbf{x}^{(k)}).$$

- $\mathbf{B}^{(k)} \propto \mathbf{I}$: steepest descent (next lecture).
- General case: variable metric (not in this course).

Shifted Newton: pseudocode

Choose $\mathbf{x}^{(0)} \in \mathcal{D}$, $\epsilon_g > 0$, $\tau_{\min} > 0$, $\gamma > 1$, and $K_{\max}$. For $k = 0, 1, \dots, K_{\max} - 1$:

- ① Compute $\mathbf{g}^{(k)} = \nabla f(\mathbf{x}^{(k)})$. Stop if $\|\mathbf{g}^{(k)}\| \leq \epsilon_g$.
- ② (Form $\mathbf{H}^{(k)} = \mathbf{H}_f(\mathbf{x}^{(k)})$.) Test $\tau = 0, \tau_{\min}, \gamma\tau_{\min}, \gamma^2\tau_{\min}, \dots$. Accept the first τ whose corresponding step satisfies

$$\mathbf{B}^{(k)} := \mathbf{H}^{(k)} + \tau \mathbf{I} \succ 0.$$

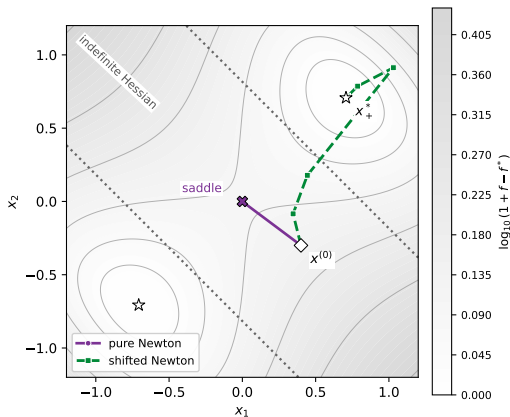
- ③ Solve the shifted Newton equation $\mathbf{B}^{(k)} \mathbf{d}_M^{(k)} = -\mathbf{g}^{(k)}$.
- ④ Take the full step $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \mathbf{d}_M^{(k)}$.

Python code: `shifted_newton(...)` in `109_newton_methods.py`.

Shifting helps avoid a saddle point

$$f(x_1, x_2) = \frac{(x_1 + x_2)^4}{16} - x_1 x_2, \quad \mathbf{x}^{(0)} = (0.4, -0.3)^T,$$

$\mathbf{0}$ is a saddle, $\mathbf{x}_{\pm}^* = \pm \frac{1}{\sqrt{2}}(1, 1)^T$ are global minimizers.



At the initial point,

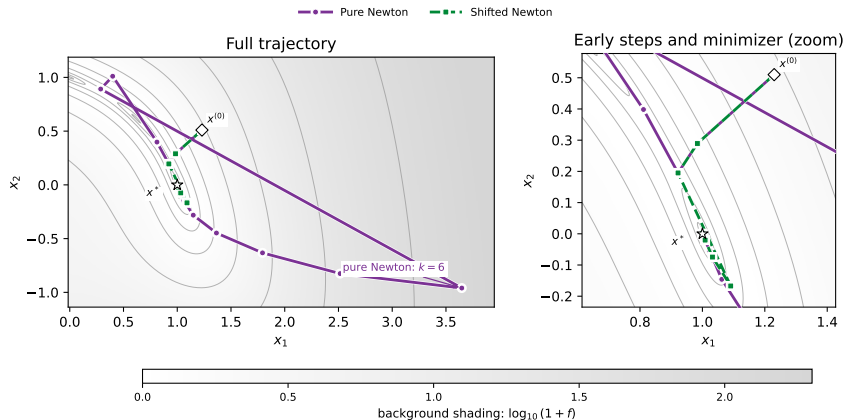
$$\lambda_{\min}(\mathbf{H}^{(0)}) = -0.985.$$

method	limit	updates
pure	saddle $\mathbf{0}$	2
shifted	$\mathbf{x}_+^*$	8

With $\tau_{\min} = 0.1$ and $\gamma = 2$, the positive shifts are 1.6, 1.6, 0.8; all later shifts are zero.

Pure and shifted Newton on the common test

Shifted Newton uses $\tau_{\min} = 0.1$, $\gamma = 2$, and the same $\mathbf{x}^{(0)} = (1.23, 0.51)^T$ as for pure Newton.



Shifted Newton on the common test

k	$f(\mathbf{x}^{(k)})$	$\ \mathbf{g}^{(k)}\ $	$\tau^{(k)}$
0	1.2865	6.740	0
1	0.070871	1.293	0
2	0.003624	0.104	0.1
3	0.014592	0.337	0
4	0.001512	0.049	0
6	8.836×10^{-7}	9.675×10^{-4}	0
7	1.288×10^{-10}	1.199×10^{-5}	0
8	2.881×10^{-18}	1.802×10^{-9}	stop

Descent directions guarantee local decrease

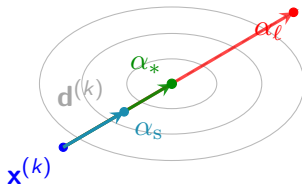
Recall that at a non-stationary $\mathbf{x}^{(k)}$, a vector $\mathbf{d}^{(k)}$ is a **descent direction** if

$$(\mathbf{g}^{(k)})^T \mathbf{d}^{(k)} < 0.$$

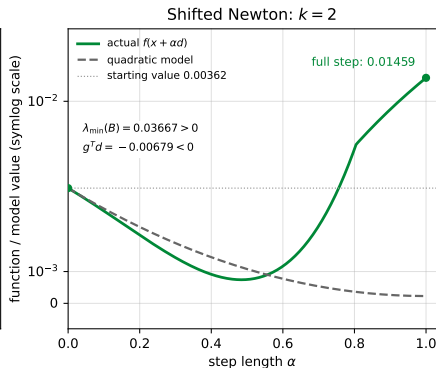
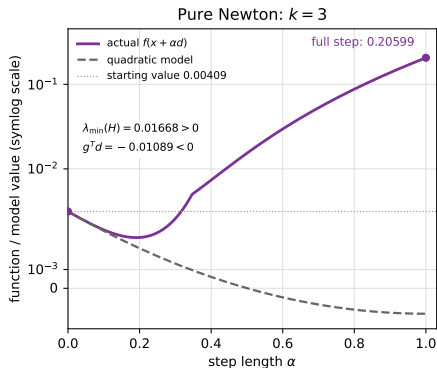
Differentiability gives

$$f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)}) = f(\mathbf{x}^{(k)}) + \alpha (\mathbf{g}^{(k)})^T \mathbf{d}^{(k)} + o(\alpha),$$

so every **sufficiently small** $\alpha > 0$ decreases f .



Descent directions can fail with a full step



- Pure Newton: $\mathbf{H}^{(3)} \succ 0$. Shifted Newton: $\mathbf{B}^{(2)} \succ 0$.
- Small steps decrease f , but $\alpha = 1$ increases f in both cases.

Exact line search

Given a descent direction $\mathbf{d}^{(k)}$, restrict f to a line:

$$\phi^{(k)}(\alpha) = f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)}).$$

Let

$$\mathcal{I}^{(k)} = \{\alpha \geq 0 : \mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)} \in \mathcal{D}\}.$$

When the minimum is attained, exact line search chooses

$$\alpha^{(k)} \in \operatorname{argmin}_{\alpha \in \mathcal{I}^{(k)}} \phi^{(k)}(\alpha).$$

NB. One-dimensional unconstrained optimization methods in Lecture 8 not necessarily guarantee global minimum.

Armijo sufficient decrease

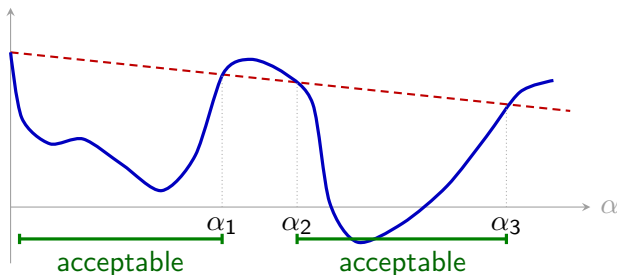
Exact line search often spends too much work on the step length. The **Armijo condition** requires only

$$f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)}) \leq f(\mathbf{x}^{(k)}) + \sigma \alpha (\mathbf{g}^{(k)})^T \mathbf{d}^{(k)},$$

where

$$0 < \sigma < 1.$$

For a descent direction, the right-hand side is below $f(\mathbf{x}^{(k)})$.



Armijo backtracking

Choose $\bar{\alpha} > 0$, $\beta \in (0, 1)$, and $\sigma \in (0, 1)$.

- ① Set $r = 0$ and test $\alpha = \bar{\alpha}$.
- ② While the point is outside $\mathcal{D}$, or the Armijo condition fails, set

$$r \leftarrow r + 1, \quad \alpha \leftarrow \beta^r \bar{\alpha}.$$

- ③ Return $\alpha^{(k)} = \alpha$.

Remark

- For Newton, use $\bar{\alpha} = 1$ so that the unit Newton step is tested first.
- For Newton, $\alpha^{(k)} = 1$ satisfies the Armijo condition ($\sigma \in (0, 1/2)$) when $\mathbf{x}^{(k)}$ is sufficiently close to a nondegenerate local minimizer $\mathbf{x}^*$.

Armijo backtracking

Choose $\bar{\alpha} > 0$, $\beta \in (0, 1)$, and $\sigma \in (0, 1)$.

- 1 Set $r = 0$ and test $\alpha = \bar{\alpha}$.
- 2 While the point is outside $\mathcal{D}$, or the Armijo condition fails, set

$$r \leftarrow r + 1, \quad \alpha \leftarrow \beta^r \bar{\alpha}.$$

- 3 Return $\alpha^{(k)} = \alpha$.

Remark

- For Newton, use $\bar{\alpha} = 1$ so that the unit Newton step is tested first.
- For Newton, $\alpha^{(k)} = 1$ satisfies the Armijo condition ($\sigma \in (0, 1/2)$) when $\mathbf{x}^{(k)}$ is sufficiently close to a nondegenerate local minimizer $\mathbf{x}^*$.
- Descent direction guarantees **finite termination** of backtracking.

Newton–Armijo: pseudocode

Choose $\mathbf{x}^{(0)} \in \mathcal{D}$, $\epsilon_g > 0$, $K_{\max}$, $\tau_{\min} > 0$, $\gamma > 1$, and $\beta, \sigma \in (0, 1)$. For $k = 0, \dots, K_{\max} - 1$:

- 1 Compute $\mathbf{g}^{(k)}$, and stop if $\|\mathbf{g}^{(k)}\| \leq \epsilon_g$. Form $\mathbf{H}^{(k)}$.
- 2 Try $\mathbf{H}^{(k)} \mathbf{d}_N^{(k)} = -\mathbf{g}^{(k)}$. If solvable and $(\mathbf{g}^{(k)})^T \mathbf{d}_N^{(k)} < 0$, set $\mathbf{d}^{(k)} = \mathbf{d}_N^{(k)}$ and $\tau^{(k)} = 0$.
- 3 **Otherwise: shifted safeguard.** Test $\tau = \tau_{\min}, \gamma\tau_{\min}, \dots$ until

$$\mathbf{B}^{(k)} = \mathbf{H}^{(k)} + \tau \mathbf{I}, \quad \mathbf{B}^{(k)} \mathbf{d}_M^{(k)} = -\mathbf{g}^{(k)}, \quad (\mathbf{g}^{(k)})^T \mathbf{d}_M^{(k)} < 0.$$

Set $\mathbf{d}^{(k)} = \mathbf{d}_M^{(k)}$ and $\tau^{(k)} = \tau$.

- 4 Apply **exact line search**, or **Armijo backtracking** starting from $\alpha = 1$, to get $\alpha^{(k)}$.
- 5 Update $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha^{(k)} \mathbf{d}^{(k)}$.

Python: `newton_armijo(...)` uses `descent_shifted_direction`.

Newton–Armijo: pseudocode

Choose $\mathbf{x}^{(0)} \in \mathcal{D}$, $\epsilon_g > 0$, $K_{\max}$, $\tau_{\min} > 0$, $\gamma > 1$, and $\beta, \sigma \in (0, 1)$. For $k = 0, \dots, K_{\max} - 1$:

- 1 Compute $\mathbf{g}^{(k)}$, and stop if $\|\mathbf{g}^{(k)}\| \leq \epsilon_g$. Form $\mathbf{H}^{(k)}$.
- 2 Try $\mathbf{H}^{(k)} \mathbf{d}_N^{(k)} = -\mathbf{g}^{(k)}$. If solvable and $(\mathbf{g}^{(k)})^T \mathbf{d}_N^{(k)} < 0$, set $\mathbf{d}^{(k)} = \mathbf{d}_N^{(k)}$ and $\tau^{(k)} = 0$.
- 3 **Otherwise: shifted safeguard.** Test $\tau = \tau_{\min}, \gamma\tau_{\min}, \dots$ until

$$\mathbf{B}^{(k)} = \mathbf{H}^{(k)} + \tau \mathbf{I}, \quad \mathbf{B}^{(k)} \mathbf{d}_M^{(k)} = -\mathbf{g}^{(k)}, \quad (\mathbf{g}^{(k)})^T \mathbf{d}_M^{(k)} < 0.$$

Set $\mathbf{d}^{(k)} = \mathbf{d}_M^{(k)}$ and $\tau^{(k)} = \tau$.

- 4 Apply **exact line search**, or **Armijo backtracking** starting from $\alpha = 1$, to get $\alpha^{(k)}$.
- 5 Update $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha^{(k)} \mathbf{d}^{(k)}$.

Python: `newton_armijo(...)` uses `descent_shifted_direction`.

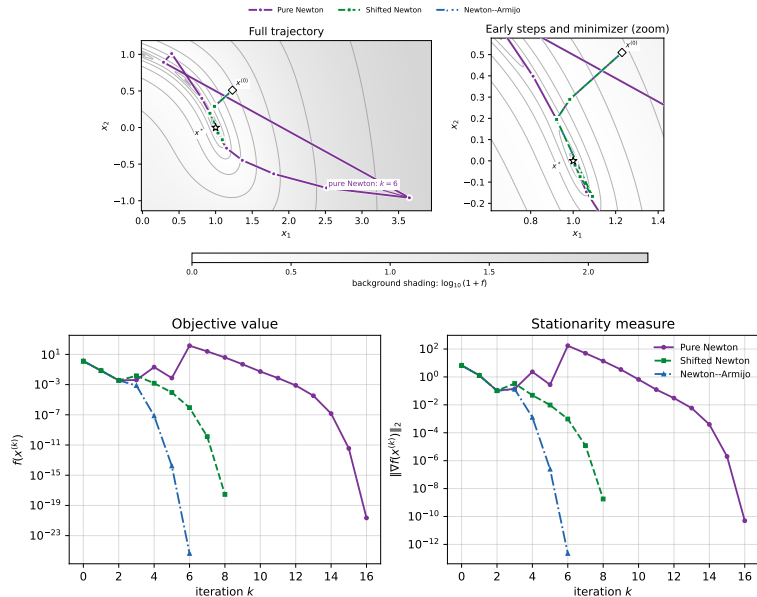
Newton–Armijo on the common test

Selected iterates:

k	$f(\mathbf{x}^{(k)})$	$\ \mathbf{g}^{(k)}\ $	$\tau^{(k)}$	$\alpha^{(k)}$
0	1.2865	6.740	0	1
1	0.070871	1.293	0	1
2	0.003624	0.104	0.1	0.5
3	0.000749	0.131	0	1
4	7.290×10^{-8}	1.274×10^{-3}	0	1
5	1.682×10^{-14}	2.471×10^{-7}	0	1
6	4.428×10^{-26}	2.271×10^{-13}	stop	

- At $k = 2$, the raw Newton direction is ascent.
- The objective is monotone. Unit steps are accepted for every $k \geq 3$.

Three methods on the common test



Three methods on the common test (cont'd)

Coupled quartic, $\mathbf{x}^{(0)} = (1.23, 0.51)^T$, and tolerance 10^{-8} .

method	updates	f eval.	g eval.	H eval.	solves
pure Newton	16	17	17	16	16
shifted Newton	8	9	9	8	8
Newton–Armijo	6	14	7	6	7

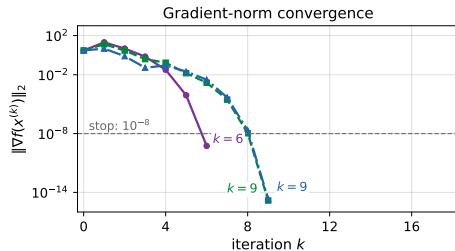
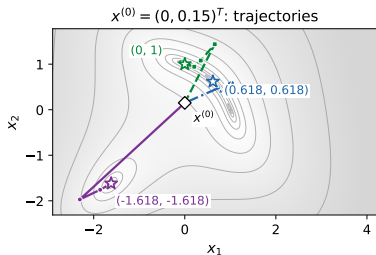
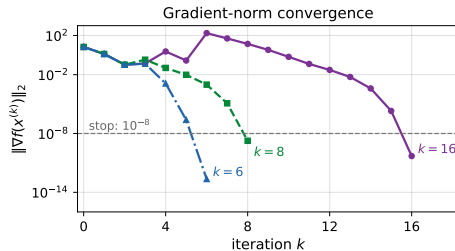
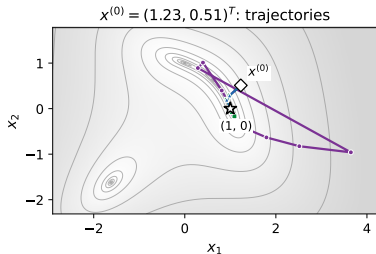
Solves include rejected candidate directions. Positive-definiteness checks for Shifted Newton are separate and are not included in this column.

Iteration counts are not total cost

The preferred method depends on the relative costs of f , ∇f , $\mathbf{H}_f$, and a linear solve.

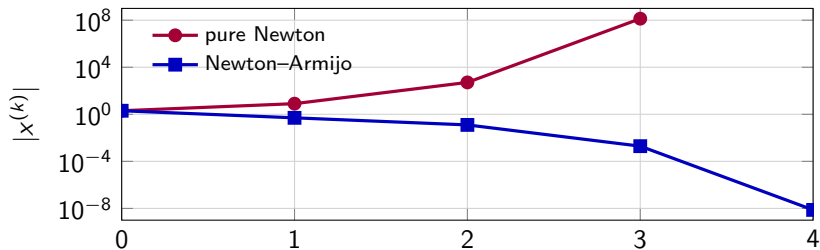
Pure Newton can be fastest

Pure Newton Shifted Newton Newton--Armijo



Pure Newton can diverge

$$f(x) = \sqrt{1+x^2}, \quad f''(x) = (1+x^2)^{-3/2} > 0, \quad x^* = 0,$$
$$d_N(x) = -x(1+x^2), \quad \alpha = 1: \quad x^{(k+1)} = -(x^{(k)})^3.$$



Unit steps

$$2 \mapsto -8 \mapsto 512 \mapsto \dots$$

The magnitudes cube and diverge.

Newton-Armijo: $\beta = 1/2, \sigma = 10^{-4}$

$$\alpha^{(0)} = \frac{1}{4}, \quad \alpha^{(k)} = 1 \quad (k \geq 1),$$
$$2 \mapsto -\frac{1}{2} \mapsto \frac{1}{8} \mapsto -\frac{1}{512} \mapsto 0.$$

Take-home messages

- **Pure Newton:** local quadratic Taylor model; not necessarily converge to local min.
- **Shifted Newton:** a positive-definite model gives a descent direction. Positive-definiteness checks (e.g., Cholesky) and linear solves.
- **Newton–Armijo:** a descent direction and sufficient decrease. Linear solves and trial function evaluations.
- **No universal winner!**

Reference proof: Newton quadratic rate (1/2)

Let $\mathbf{e}^{(k)} = \mathbf{x}^{(k)} - \mathbf{x}^*$, $\mathbf{H}^{(k)} = \mathbf{H}_f(\mathbf{x}^{(k)})$, and $\mathbf{H}_* = \mathbf{H}_f(\mathbf{x}^*)$. Since $\nabla f(\mathbf{x}^*) = \mathbf{0}$,

$$\nabla f(\mathbf{x}^{(k)}) = \left(\int_0^1 \mathbf{H}_f(\mathbf{x}^* + t\mathbf{e}^{(k)}) dt \right) \mathbf{e}^{(k)}.$$

The Newton update gives

$$\mathbf{e}^{(k+1)} = (\mathbf{H}^{(k)})^{-1} \left(\int_0^1 [\mathbf{H}^{(k)} - \mathbf{H}_f(\mathbf{x}^* + t\mathbf{e}^{(k)})] dt \right) \mathbf{e}^{(k)}.$$

Nonsingularity of $\mathbf{H}_*$ and continuity imply, in a sufficiently small neighborhood,

$$\|(\mathbf{H}^{(k)})^{-1}\|_2 \leq 2\|\mathbf{H}_*^{-1}\|_2.$$

Reference proof: Newton quadratic rate (2/2)

Local Hessian Lipschitz continuity gives

$$\|\mathbf{H}^{(k)} - \mathbf{H}_f(\mathbf{x}^* + t\mathbf{e}^{(k)})\|_2 \leq L(1-t)\|\mathbf{e}^{(k)}\|.$$

Hence

$$\begin{aligned}\|\mathbf{e}^{(k+1)}\| &\leq 2\|\mathbf{H}_*^{-1}\|_2 \int_0^1 L(1-t) dt \|\mathbf{e}^{(k)}\|^2 \\ &= L\|\mathbf{H}_*^{-1}\|_2 \|\mathbf{e}^{(k)}\|^2.\end{aligned}$$

Let $M = L\|\mathbf{H}_*^{-1}\|_2$. Choose a ball inside the neighborhood with radius δ satisfying $M\delta \leq 1/2$. Then

$$\|\mathbf{e}^{(k)}\| \leq \delta \implies \|\mathbf{e}^{(k+1)}\| \leq M\|\mathbf{e}^{(k)}\|^2 \leq \frac{1}{2}\|\mathbf{e}^{(k)}\|.$$

Induction keeps the iterates in the ball, gives $\mathbf{e}^{(k)} \rightarrow \mathbf{0}$, and preserves the quadratic error bound.

Eventual acceptance of the unit step

Assume $\mathbf{x}^{(k)} \rightarrow \mathbf{x}^*$, the Hessian is locally Lipschitz near $\mathbf{x}^*$, and $\mathbf{H}_f(\mathbf{x}^*) \succ 0$. Eventually $\mathbf{H}^{(k)} \succ 0$, so the shifted safeguard is inactive and the algorithm uses Newton directions. Taylor expansion gives

$$\begin{aligned} f(\mathbf{x}^{(k)} + \mathbf{d}_N^{(k)}) &= f(\mathbf{x}^{(k)}) + (\mathbf{g}^{(k)})^T \mathbf{d}_N^{(k)} + \frac{1}{2}(\mathbf{d}_N^{(k)})^T \mathbf{H}^{(k)} \mathbf{d}_N^{(k)} + o(\|\mathbf{d}_N^{(k)}\|^2) \\ &= f(\mathbf{x}^{(k)}) + \frac{1}{2}(\mathbf{g}^{(k)})^T \mathbf{d}_N^{(k)} + o(\|\mathbf{d}_N^{(k)}\|^2), \end{aligned}$$

because $\mathbf{H}^{(k)} \mathbf{d}_N^{(k)} = -\mathbf{g}^{(k)}$. Uniform positive definiteness implies $-(\mathbf{g}^{(k)})^T \mathbf{d}_N^{(k)} \geq c\|\mathbf{d}_N^{(k)}\|^2$ nearby. Hence, for $0 < \sigma < 1/2$, the unit step satisfies the Armijo condition for every sufficiently large k .